

Indian Job Market Analysis | 2025

Project Report

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Tools Used: Python | SQL | Power BI

Project Type: Data Analytics Portfolio Project

1. Executive Summary

This report documents the Indian Job Market Analysis | 2025 project, a data analytics study of 97,929 job listings across the Indian market. The project follows a full analytics pipeline — data cleaning and exploratory analysis in Python, structured querying in SQL, and interactive visualization in Power BI — to convert raw job posting data into insights on hiring demand, company activity, and salary trends.

Key findings include an average salary of 318.61K across 18,328 companies, with Bengaluru emerging as the leading hiring hub (16.8K job postings) and Application Developer as the most in-demand role (2,039 postings). Salary analysis further shows a clear, expected progression: compensation rises with experience level from Fresher to Mid-Level to Senior.

The sections below cover the project's objectives, dataset, methodology, and findings in detail, followed by challenges encountered, current limitations, future enhancements, and the skills it demonstrates.

2. Project Overview

The Indian Job Market Analysis | 2025 project is a comprehensive data analytics project designed to analyze employment trends, hiring patterns, salary insights, and workforce demand across different job roles, companies, locations, and experience levels.

The project aims to transform raw job listing data into meaningful business insights by applying data cleaning, exploratory data analysis, SQL-based analysis, and interactive dashboard development.

Using Python, SQL, and Power BI, this analysis identifies key market trends such as high-demand job roles, leading hiring locations, top recruiting companies, salary distribution patterns, and the relationship between experience levels and compensation.

The final Power BI dashboard provides an interactive and user-friendly platform for exploring job market insights and supporting data-driven career and recruitment decisions.

3. Business Objective

The primary objective of this project is to analyze the Indian job market landscape and uncover valuable insights related to employment opportunities, hiring trends, and salary patterns.

The analysis focuses on:

- Identifying the most in-demand job roles based on job posting frequency

- Understanding major hiring locations and regions with higher employment opportunities
- Analyzing top companies contributing to job creation in the market
- Evaluating salary trends across different job roles and experience levels
- Comparing compensation patterns between fresher, mid-level, and senior professionals
- Building an interactive Power BI dashboard to present insights in a clear and effective manner

The project helps demonstrate how data analytics can support better decision-making for job seekers, recruiters, and business stakeholders.

4. Dataset Description

The dataset used for this project contains job listing information collected from the Indian job market. It includes details related to job roles, companies, locations, salary ranges, experience requirements, and job posting information.

The dataset provides valuable information for analyzing employment trends and understanding the demand for different skills and job positions.

Dataset Details

Attribute	Value
Dataset Name	Indian Job Market Dataset
Total Records	97,929 job listings
Total Columns	17+ attributes after data processing

Key Features

- Job Information: Job title, job ID, job description, and posting details
- Company Information: Company name and company identifiers
- Location Details: City-wise job availability and hiring trends
- Salary Information: Minimum salary, maximum salary, salary categories, and salary ranges
- Experience Information: Required experience levels and categorized experience groups
- Company Reviews: Review count and company rating information

The dataset was analyzed and transformed to extract meaningful insights about hiring demand, salary patterns, and employment trends in the Indian job market.

5. Data Cleaning & Preparation

Before performing analysis, the raw dataset was cleaned and prepared to improve data quality, consistency, and accuracy.

Data Preprocessing Steps

Handling Missing Values

Missing values in important columns such as salary details, company ratings, and review counts were identified and handled appropriately.

Removing Duplicate Records

Duplicate job listings were detected and removed to maintain data accuracy and avoid repeated analysis.

Data Type Conversion

Columns were converted into appropriate data types, including numerical, categorical, and date formats for efficient analysis.

Salary Data Processing

Salary-related columns were cleaned and structured to enable salary comparison and analysis across job roles and experience levels.

Experience Level Categorization

Job experience requirements were transformed into meaningful categories such as Fresher, Mid-Level, and Senior.

Feature Preparation

Additional analytical columns such as Salary Category and Experience Level were created to support deeper insights in SQL analysis and Power BI visualization.

After the cleaning process, the final dataset was ready for exploratory analysis, SQL querying, and dashboard development.

6. Tools & Technologies Used

This project was developed using a combination of data analysis, database management, and visualization tools to perform end-to-end analytics.

Programming & Data Analysis — Python

- Used for data cleaning, preprocessing, exploratory data analysis, and generating analytical insights

Libraries used: Pandas, NumPy, Matplotlib, Seaborn

Database & Query Analysis — SQL (SQLite)

- Used for storing processed data and performing analytical queries

Key analysis included: hiring trends, company-wise job distribution, salary analysis, experience-based comparisons

Data Visualization — Power BI

- Used to create an interactive dashboard for presenting job market insights

Developed visualizations for: job demand analysis, hiring locations, company analysis, salary trends, experience-level comparisons

Development Environment

- Jupyter Notebook
- Visual Studio Code

- DB Browser for SQLite

The combination of these tools enabled a complete data analytics workflow, from raw data processing to interactive business intelligence reporting.

7. Exploratory Data Analysis (Python)

Python was used to perform exploratory data analysis (EDA) and understand the structure, patterns, and key characteristics of the job market dataset. The analysis included data inspection, statistical analysis, and visualization to identify important trends and relationships.

Key EDA Activities

Dataset Exploration

Analyzed dataset structure, dimensions, column information, and data distributions to understand available job market attributes.

Data Quality Analysis

Checked missing values, duplicate records, and inconsistencies to ensure reliable analysis.

Job Demand Analysis

Analyzed job titles to identify the most frequently posted and highly demanded job roles.

Company Hiring Analysis

Evaluated company-wise job postings to identify organizations with higher hiring activity.

Location Analysis

Analyzed job distribution across different cities to identify major employment hubs.

Salary Analysis

Studied salary ranges, salary categories, and compensation patterns across different roles.

Experience Analysis

Compared job opportunities and salary trends across Fresher, Mid-Level, and Senior experience categories.

The insights generated through Python analysis helped in designing meaningful SQL queries and Power BI dashboard visualizations.

8. SQL Analysis

SQL was used to perform structured analysis on the cleaned job market dataset. The data was stored in an SQLite database, and multiple analytical queries were executed to extract meaningful business insights.

Key SQL Analysis Performed

Job Market Overview

Calculated total job listings, total companies, and overall dataset statistics.

Company Hiring Analysis

Analyzed company-wise job postings to identify organizations with the highest hiring activity.

Location-Based Analysis

Identified top hiring locations based on the number of available job opportunities.

Salary Analysis

Analyzed minimum and maximum salary ranges to understand compensation patterns across job categories.

Experience-Level Analysis

Compared salary trends and job availability among Fresher, Mid-Level, and Senior professionals.

Job Role Analysis

Identified high-demand roles and analyzed salary trends for different positions.

The SQL analysis provided structured insights that were later visualized through Power BI dashboards for better understanding and decision-making.

Sample Query

Example of a query used to compare salary by experience level:

```
SELECT ExperienceLevel, AVG(maximumSalary) AS Avg_Salary, COUNT(jobId) AS Job_Count
FROM job_listings
GROUP BY ExperienceLevel
ORDER BY Avg_Salary DESC;
```

9. Power BI Dashboard Development

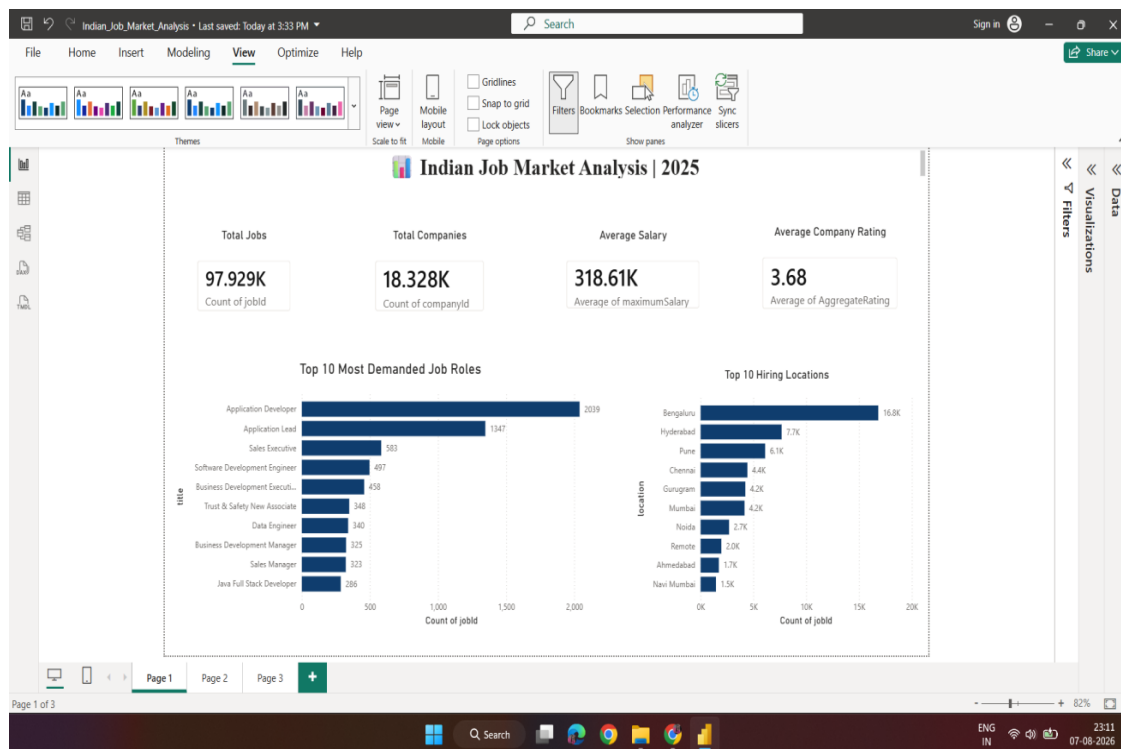
Power BI was used to create an interactive dashboard that presents key insights from the Indian Job Market Analysis project. The dashboard was designed to provide a clear overview of employment trends, hiring patterns, and salary insights through interactive visualizations.

Page 1: Indian Job Market Analysis | 2025

This page provides a high-level overview of the job market with key performance indicators and demand analysis.

Visualizations included:

- Total Job Listings
- Total Companies
- Average Salary
- Average Company Rating
- Top 10 Most Demanded Job Roles
- Top 10 Hiring Locations

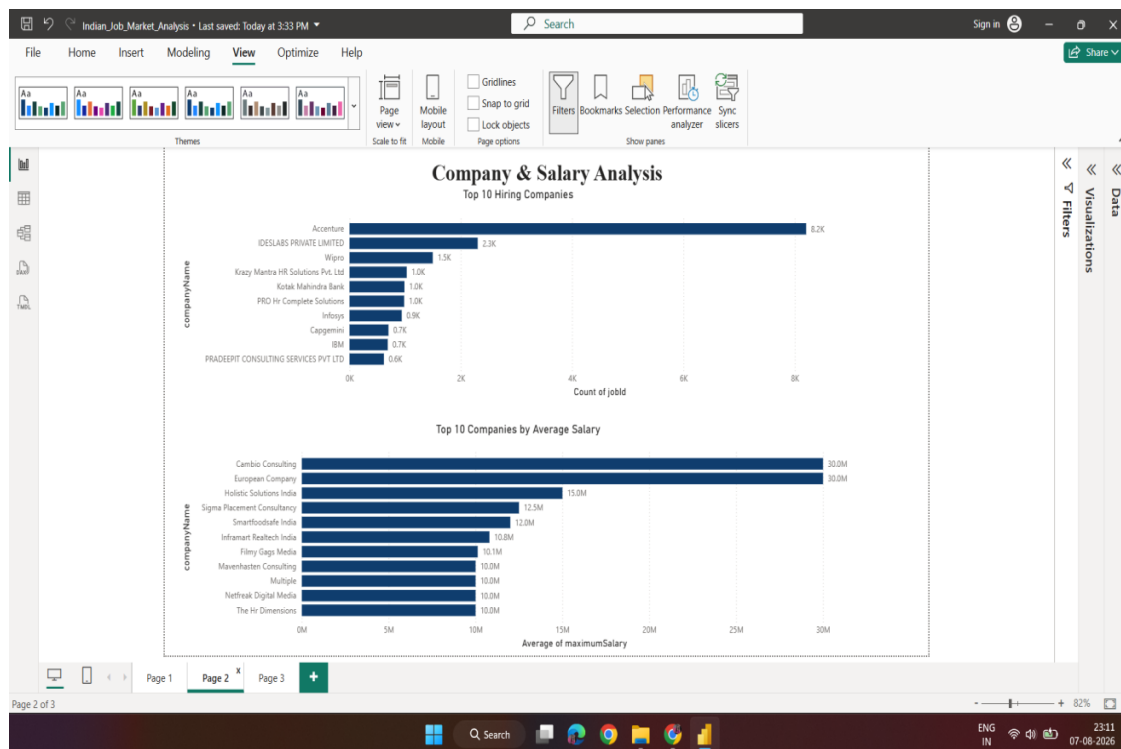


Page 2: Company & Salary Analysis

This page focuses on company-level hiring activity and salary comparison.

Visualizations included:

- Top 10 Hiring Companies
- Top 10 Companies by Average Salary

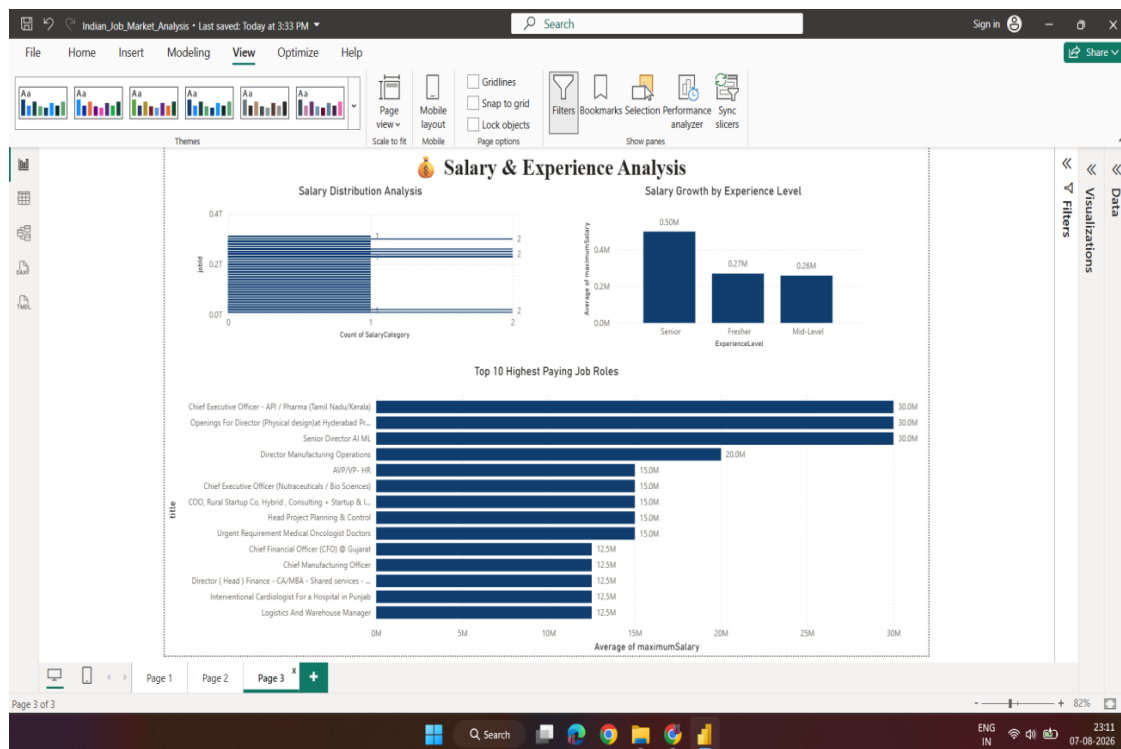


Page 3: Salary & Experience Analysis

This page analyzes salary patterns across different categories and experience levels.

Visualizations included:

- Salary Distribution Analysis
- Salary Growth by Experience Level
- Top 10 Highest Paying Job Roles



The dashboard enables users to interact with the data and explore job market trends through a visually engaging and user-friendly interface.

10. Key Insights

The analysis of the Indian job market dataset revealed several important trends related to hiring demand, company activity, salary patterns, and experience requirements.

Hiring Trends

- The analysis identified the most demanded job roles based on the number of job postings, highlighting positions with higher market demand
- Technology-focused roles showed significant hiring activity, indicating strong demand for skilled professionals in the IT sector

Location Insights

- Major cities such as Bengaluru, Hyderabad, Pune, Chennai, and Gurugram emerged as leading hiring locations
- These locations represent important employment hubs with higher concentrations of job opportunities

Company Insights

- Several companies contributed significantly to job creation and hiring activity
- Company-wise analysis helped identify organizations with higher recruitment demand

Salary Insights

- Salary analysis showed variations across different job roles and experience levels
- Senior-level positions generally offered higher compensation compared to fresher and mid-level roles

Experience-Based Insights

- Experience level had a direct impact on salary growth
- Professionals with higher experience requirements received better salary opportunities compared to entry-level roles

These insights demonstrate how data analytics can be used to understand market trends and support informed career and recruitment decisions.

11. Challenges Faced & Solutions

Cleaning a large, inconsistent real-world dataset

With nearly 98,000 job listings sourced from varied postings, salary fields, ratings, and review counts had missing or inconsistent entries that needed careful handling before any reliable analysis could begin.

Standardizing experience levels

Raw experience requirements were free-text and inconsistent, so they had to be mapped into clear categories (Fresher, Mid-Level, Senior) to make salary and demand comparisons meaningful.

Keeping Python, SQL, and Power BI in sync

Since the same cleaned dataset fed all three stages, any adjustment made during EDA in Python had to be carried through consistently to the SQLite database and the Power BI model.

Designing a multi-page dashboard that stays focused

With job, company, location, salary, and experience dimensions all in play, structuring the dashboard into three focused pages (Overview, Company & Salary, Salary & Experience) kept each page readable instead of overcrowded.

12. Limitations

- The dataset represents a snapshot of job postings rather than a continuously updated live feed
- Salary figures reflect listed ranges from job postings, which may not always match actual offered or accepted compensation
- The analysis does not account for factors like company size, industry sector, or cost-of-living differences across cities
- Experience-level categorization is based on posting requirements, not verified candidate experience

13. Future Enhancements

- Incorporate industry/sector classification to compare hiring and salary trends across sectors, not just roles

- Add skill-based analysis (e.g., most in-demand technical skills per role) using job description text
- Build a salary prediction model based on role, location, experience, and company
- Automate data refresh so the Power BI dashboard reflects new job postings over time
- Publish the dashboard to Power BI Service for shareable, web-based access
- Add city-level cost-of-living adjustment to make salary comparisons more meaningful across locations

14. Skills Demonstrated

- Data cleaning and preprocessing on a large real-world dataset using Python (Pandas, NumPy)
- Exploratory data analysis and visualization using Matplotlib and Seaborn
- Database management and business-focused SQL querying (SQLite)
- Dashboard design and interactive data visualization in Power BI
- Translating raw employment data into hiring, salary, and experience-based insights
- End-to-end project documentation and reporting

15. Conclusion

The Indian Job Market Analysis | 2025 project demonstrates the complete data analytics workflow, from data cleaning and exploration to SQL analysis and interactive dashboard development.

By using Python, SQL, and Power BI, the project successfully transformed raw job listing data into meaningful insights about hiring trends, salary patterns, company performance, and experience-based opportunities.

The analysis provides a clear understanding of the current job market landscape and highlights how data-driven approaches can support better decision-making for job seekers, recruiters, and organizations. This project enhanced practical skills in data preprocessing, exploratory analysis, database querying, business intelligence reporting, and data visualization, making it a valuable addition to a Data Analyst portfolio.

16. References & Data Source

- Dataset: Indian Job Market Dataset (97,929 job listings)
- Tools used: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL (SQLite), Power BI
- Development environment: Jupyter Notebook, Visual Studio Code, DB Browser for SQLite